Sine, cosine, and tangent ratios

Core Concepts: Sine, cosine, and tangent ratios

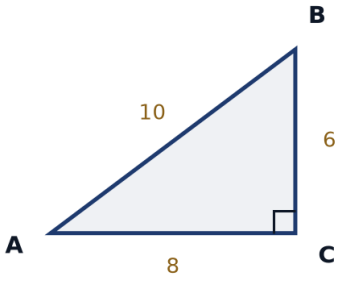
1 For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(2)$.

a. 17

b. 23

c. 12

d. 21



2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(3)$.

a. 23

b. 29

c. 18

d. 27

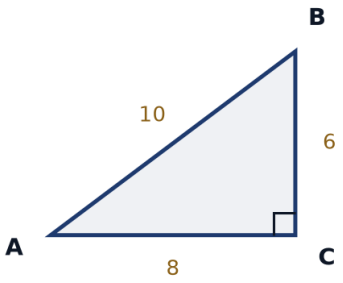
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(4)$.

a. 29

b. 35

c. 24

d. 33

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8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

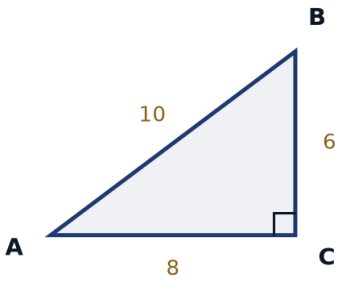
9 A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5



Exam-Style Problem Solving & Modeling

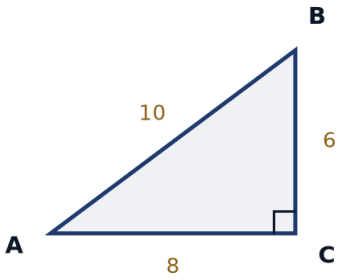
10 For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(5)$.

a. 35

b. 41

c. 30

d. 39



11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

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12 A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

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